8 **(a)** Fig. 8.1 shows a simplified diagram of an a.c. generator.

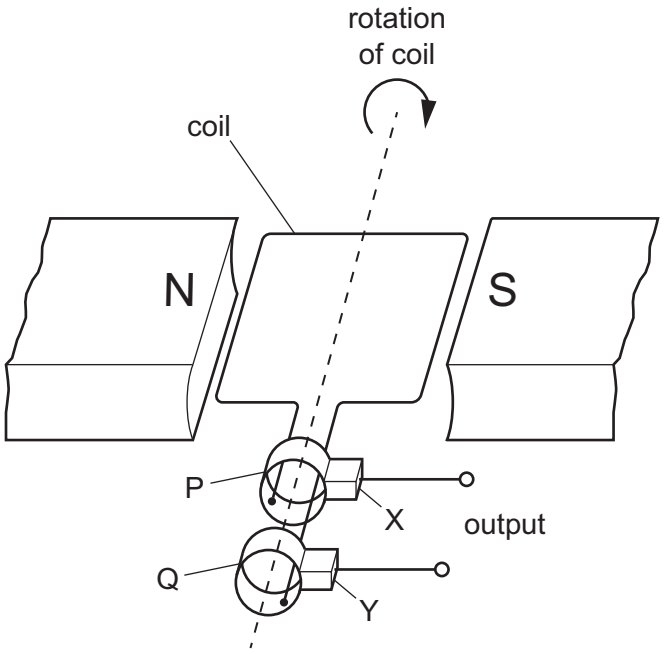

Fig. 8.1

(i) State the names of components:

P and Q .....

X and Y. .....
[2]

(ii) Explain why an electromotive force (e.m.f.) is only induced when the coil is turning.

.....

..... [1]

(iii) State **one** possible change that causes a larger e.m.f. to be induced.

..... [1]

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(b) Fig. 8.2 shows a circuit diagram.

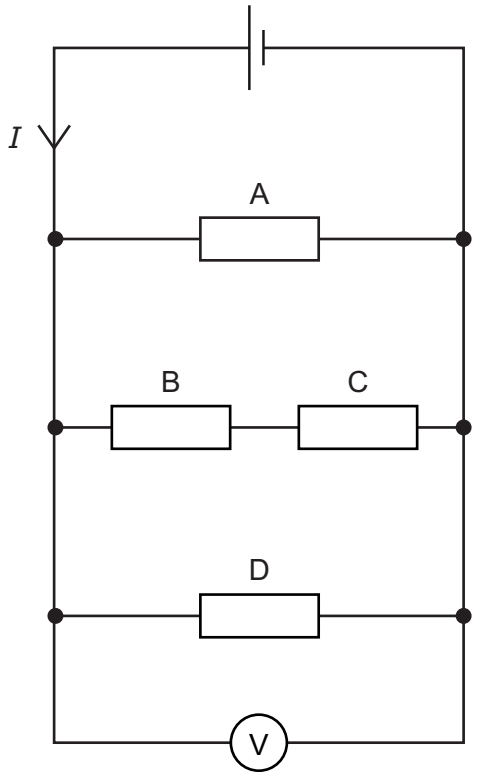

Fig. 8.2

Resistors A, B, C and D are identical. The current in resistor A is 2.4 A.

(i) State the value of the current in resistor B. Explain your answer.

current in resistor B .....

explanation .....

.....

[2]

(ii) Calculate the value of the current I.

current I = ... [1]

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(iii) The reading on the voltmeter is 5.0 V.

Calculate the resistance of resistor A.

resistance = ... [2]

[Total: 9]

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